

Definition 12.1

A *transposition* in S_n is a cycle (i_1, i_2) of length 2.

Theorem 12.2

Every permutation in S_n can be written as a product of transpositions.

Proof. By Theorem 11.5 every permutation is product of cycles, so it is enough to show that every cycle can be written as a product of transpositions. This is true, since if $(i_1, i_2, \dots, i_r)$ is a cycle in S_n then

$$(i_1, i_2, \dots, i_r) = (i_1, i_r) \circ (i_1, i_{r-1}) \circ \cdots \circ (i_1, i_2)$$

□

Note. A permutation can be written as a product of cycles in many different ways:

$$\begin{aligned} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{bmatrix} &= (1, 3) \circ (1, 2) \\ &= (2, 3) \circ (1, 3) \\ &= (1, 3) \circ (4, 2) \circ (1, 2) \circ (1, 4) \\ &= (2, 4) \circ (1, 2) \circ (2, 3) \circ (1, 4) \\ &= \dots \end{aligned}$$

Theorem 12.3

Let $\alpha \in S_n$ and let $\alpha = \beta_1 \circ \beta_2 \circ \cdots \circ \beta_r$ be a decomposition of α into a product of r transpositions.

- If r is even, then every other decomposition of α into transpositions consists of an even number of transpositions.
- If r is odd, then every other decomposition of α into transpositions consists of an odd number of transpositions.

Lemma 12.4

Let $\beta_1, \dots, \beta_r$ be transpositions in S_n such that

$$\beta_1 \circ \beta_2 \circ \dots \circ \beta_r = \varepsilon$$

where ε is the identity permutation. Then r is an even number.

Proof. We will prove by induction with respect to k the following statement:

For any $k \geq 2$, if $\beta_1 \circ \beta_2 \circ \dots \circ \beta_r = \varepsilon$ and $r \leq k$ then r is an even number.

If $k = 2$ this holds, since the only way to write ε as a product of 1 or 2 transpositions is $\beta \circ \beta^{-1}$, which uses 2 transpositions.

For the inductive step, assume then that the statement holds for some k . We need to show that it also holds for $k + 1$. Let then $\beta_1, \dots, \beta_r$ be transpositions such that $r \leq k + 1$ and

$$\beta_1 \circ \beta_2 \circ \dots \circ \beta_r = \varepsilon \quad (*)$$

Assume that one of the transpositions β_i is of the form (a, b) for some $a, b \in \{1, \dots, n\}$. One can check that the following identities hold:

- $(a, b) \circ (c, d) = (c, d) \circ (a, b)$
- $(a, b) \circ (b, c) = (b, c) \circ (a, c)$

Here a, b, c, d are distinct elements of the set $\{1, \dots, n\}$. These identities say that when multiplying transpositions, we can move the transposition involving a toward the right side without changing the number of transpositions. Using this observation, we can rewrite the equation $(*)$ as follows:

$$\gamma_1 \circ \dots \circ \gamma_{r-k} \circ (a, b_1) \circ (a, b_2) \circ \dots \circ (a, b_k) = \varepsilon \quad (**)$$

where $\gamma_1, \dots, \gamma_{r-k}$ are transpositions that do not involve a . If $b_1 \neq b_i$ for $i = 2, \dots, k$, then the permutation on the left hand side of the equation $(**)$ would send b_1 to a , which is impossible. This means that there is $i > 1$ such that $b_1 \neq b_2, \dots, b_{i-1}$ and $b_1 = b_i$. We will need one more identity:

- $(a, b) \circ (a, c) = (b, c) \circ (a, b)$ for distinct elements a, b, c .

Using this identity, we can bring the equation $(**)$ to the following form:

$$\gamma_1 \circ \dots \circ \gamma_{r-k} \circ (b_1, b_2) \circ \dots \circ (b_1, b_{i-1}) \circ (a, b_1) \circ (a, b_i) \circ (a, b_{i+1}) \circ \dots \circ (a, b_n) = \varepsilon$$

Since $b_1 = b_i$ we have $(a, b_1) \circ (a, b_i) = \varepsilon$, and the above equation becomes

$$\gamma_1 \circ \dots \circ \gamma_{r-k} \circ (b_1, b_2) \circ \dots \circ (b_1, b_{i-1}) \circ (a, b_{i+1}) \circ \dots \circ (a, b_n) = \varepsilon$$

This expresses ε as a product of $r - 2$ transpositions. Since $r \leq k + 1$, thus $r - 2 \leq k$, and so, by the inductive assumption, $r - 2$ must be an even number. Therefore r is an even as well. $\square$

Proof of Theorem 12.3. Assume that a permutation α can be written as a product of transpositions in two different ways:

$$\alpha = \beta_1 \circ \beta_2 \circ \dots \circ \beta_r$$

$$\alpha = \gamma_1 \circ \gamma_2 \circ \dots \circ \gamma_s$$

Then we have

$$\begin{aligned} \varepsilon &= \alpha \circ \alpha^{-1} \\ &= (\beta_1 \circ \beta_2 \circ \dots \circ \beta_r) \circ (\gamma_1 \circ \gamma_2 \circ \dots \circ \gamma_s)^{-1} \\ &= (\beta_1 \circ \beta_2 \circ \dots \circ \beta_r) \circ (\gamma_s^{-1} \circ \gamma_{s-1}^{-1} \circ \dots \circ \gamma_1^{-1}) \\ &= (\beta_1 \circ \beta_2 \circ \dots \circ \beta_r) \circ (\gamma_s \circ \gamma_{s-1} \circ \dots \circ \gamma_1) \end{aligned}$$

This means that ε is a product of $r + s$ transpositions. Since by Lemma 12.4, $r + s$ is an even number, thus either both r and s are even numbers or they are both odd. $\square$

Definition 12.5

A permutation $\alpha \in S_n$ is *even* if it can be written as a product of even number of transpositions and it is *odd* if it can be written as a product of an odd number of transpositions.

Theorem 12.6

The subset of S_n consisting of all even permutations is a subgroup of S_n .

Definition 12.7

The subgroup of S_n consisting of even permutations is called an *alternating group on n letters* and it is denoted by A_n

Theorem 12.8

For $n \geq 2$ the alternating group A_n has order $\frac{n!}{2}$.

Proof. Let B_n be the set of all odd permutations in S_n . Since $|S_n| = n!$, it is enough to show that $|A_n| = |B_n|$, i.e. that there exists a bijection $f: A_n \rightarrow B_n$. Such bijection can be defined by $f(\alpha) = (1, 2) \circ \alpha$. $\square$

Definition 12.9

The *sign* of a permutation $\alpha \in S_n$ is defined as follows:

$$\text{sign}(\alpha) = \begin{cases} +1 & \text{if } \alpha \text{ is even} \\ -1 & \text{if } \alpha \text{ is odd} \end{cases}$$

Note. Recall that for a square matrix A we can compute its determinant $\det A$. The determinant can be defined using permutations and their signs as follows. For a matrix

$$A = \begin{bmatrix} a_{1,1} & \dots & a_{1,n} \\ \vdots & & \vdots \\ a_{n,1} & \dots & a_{n,n} \end{bmatrix}$$

we set

$$\det A = \sum_{\alpha \in S_n} \text{sign}(\alpha) \cdot a_{1,\alpha(1)} \cdot a_{2,\alpha(2)} \cdot \dots \cdot a_{n,\alpha(n)}$$